



Simulation & Modelling

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Imagination at work.

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Simulation & Modelling

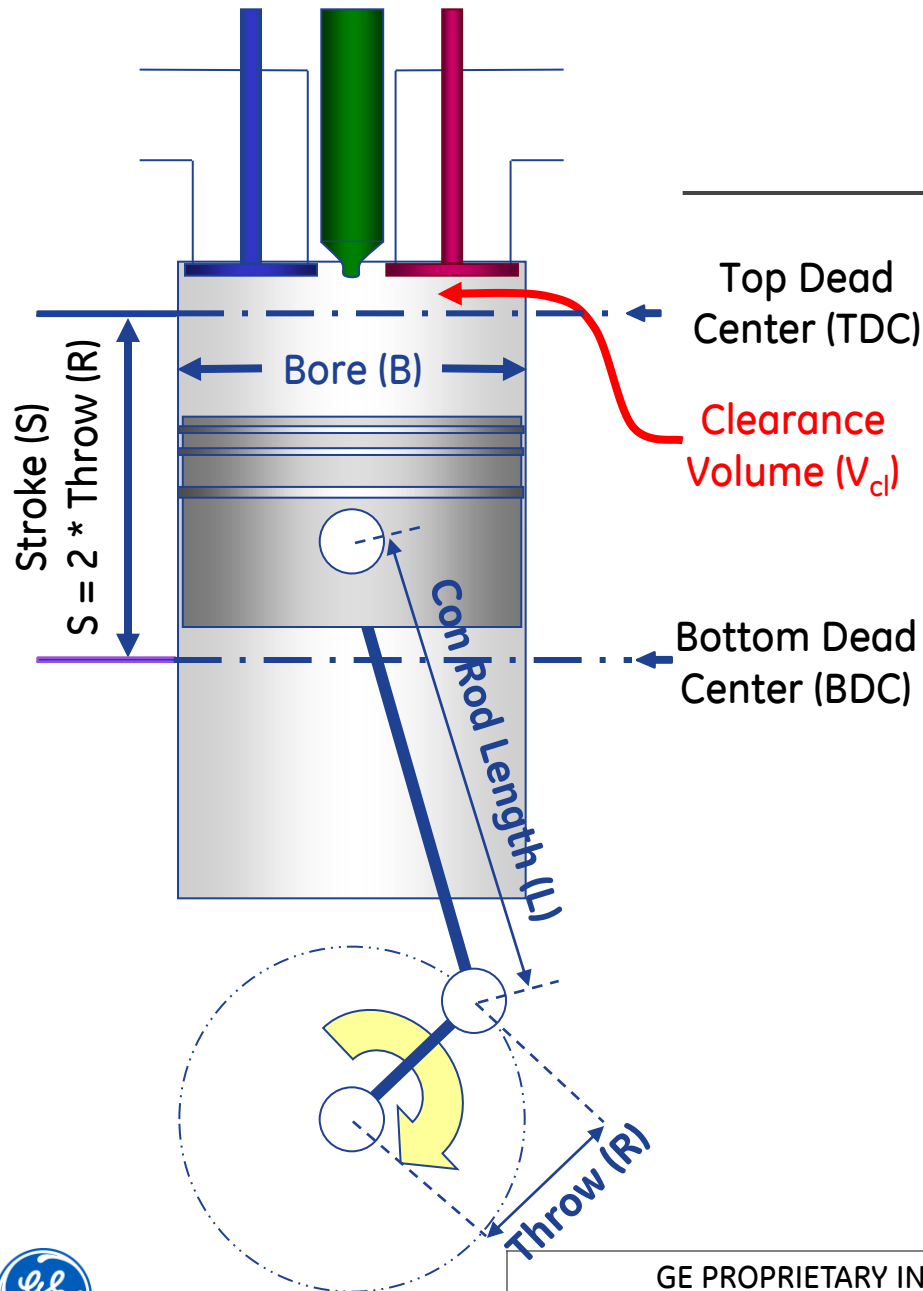


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Performance basics

Basic engine geometry



Displacement (V_{disp})

$$V_{disp} = \left[\frac{\pi * B^2}{4} \right] * S$$

Compression Ratio (CR)

$$CR = \frac{V_{max}}{V_{min}} = \frac{V_{BDC}}{V_{TDC}}$$

$$CR = \frac{V_{disp} + V_{cl}}{V_{cl}}$$

$$V_{cl} = \frac{V_{disp}}{CR - 1}$$

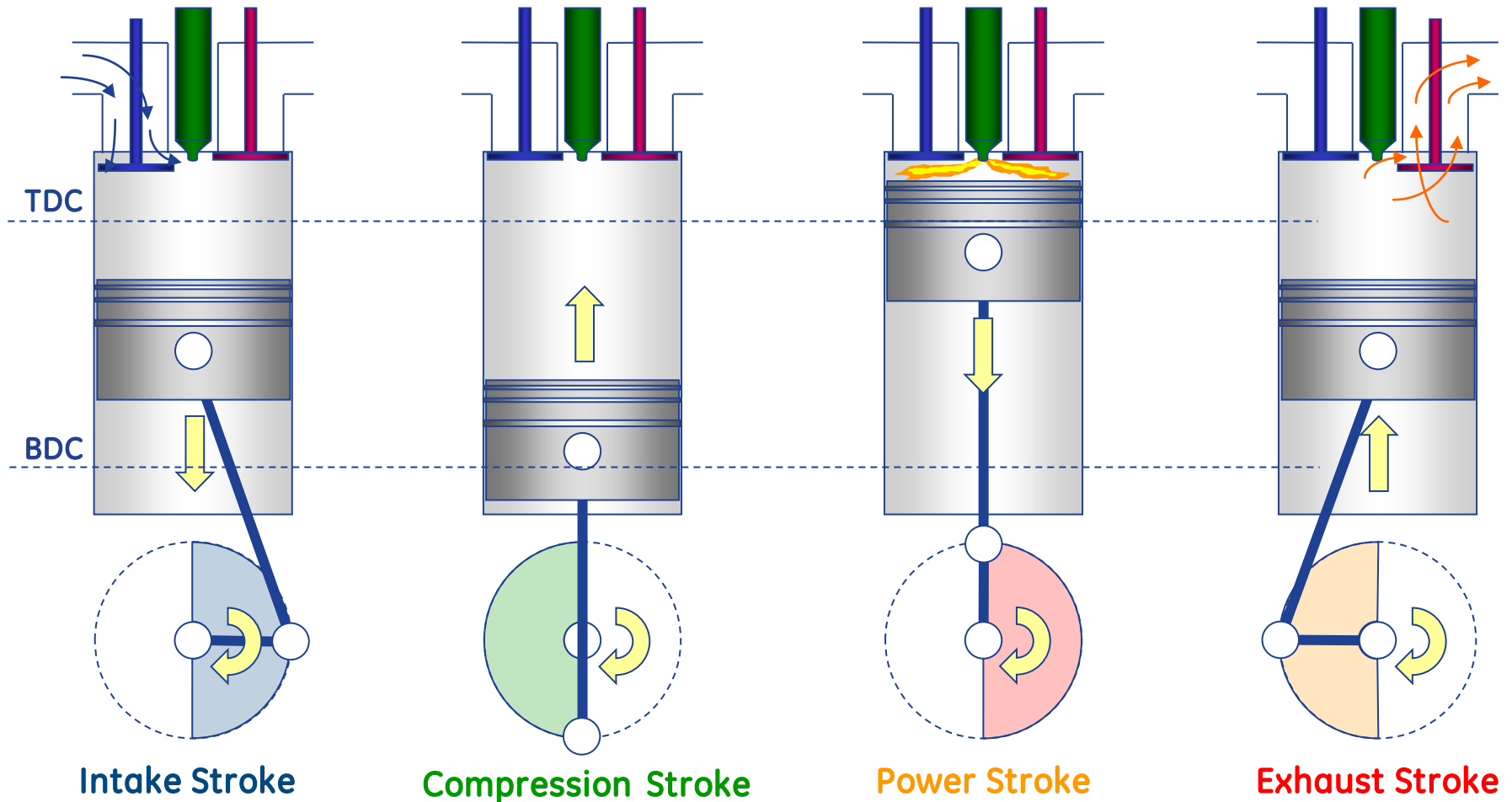


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Performance basics

Four stroke cycle

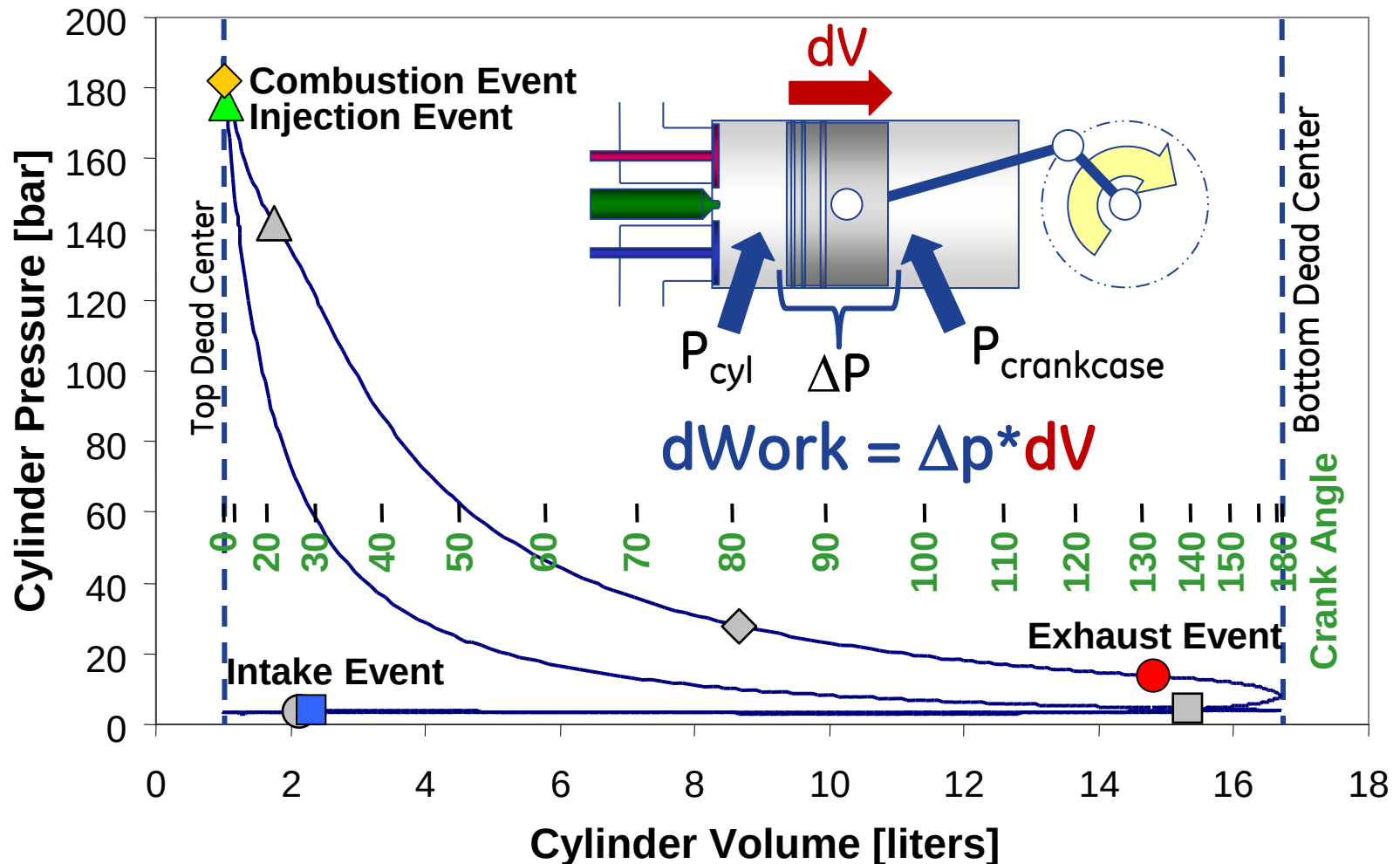


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Performance basics

Four stroke cycle

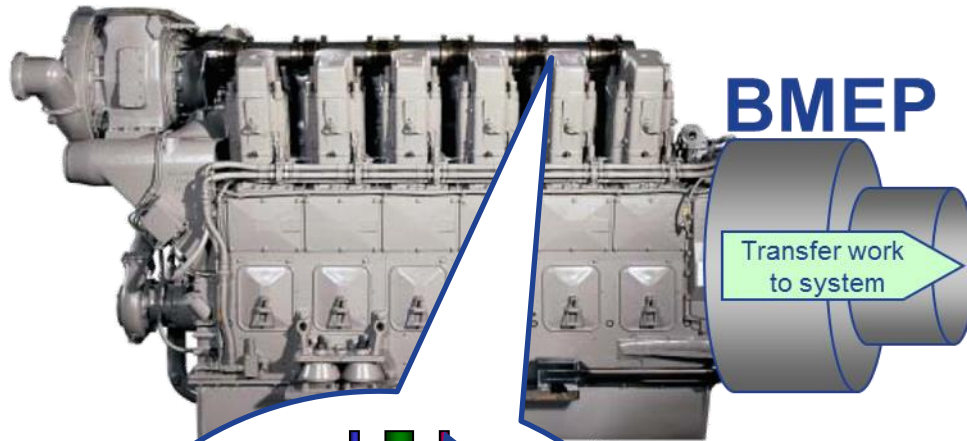


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Performance basics

Indicated mean effective pressure – brake mean effective pressure



BMEP

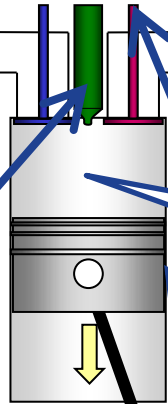
Transfer work
to system

$$BMEP = NIMEP - FMEP$$

$$FMEP = NIMEP - BMEP$$

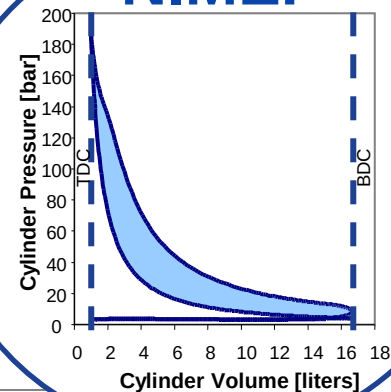
Losses occur in transferring
indicated work to the
point of engine output (brake)

Parasitic



Friction

NIMEP



Interaction between
Gas and piston



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Performance basics

Reciprocating engines equations – bmep equivalent to torque

$$\text{bmep} = \text{nimep} - \text{fmep}$$

$$\text{Brake Power} = \frac{\text{Displacement} \cdot \text{speed} \cdot \text{bmep}}{\frac{\text{revolutions}}{\text{cycle}}} = \text{Torque} \cdot \text{rotational speed}$$

$$P_b = \frac{D \cdot n \cdot \text{bmep}}{\frac{\text{revolutions}}{\text{cycle}}} = T \cdot \omega \quad \frac{\text{revolutions}}{\text{cycle}} = 2 \text{ (four stroke)}$$

$$P_b = \frac{D \cdot n \cdot \text{bmep}}{2} = T \cdot 2 \pi n$$

$$\text{bmep} = \frac{4\pi T}{D}$$

$$P_b = \frac{D \cdot n \cdot \text{bmep}}{2}$$



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Performance basics

Reciprocating engines equations – efficiency, Lambda

$$\text{brake specific fuel consumption} = \frac{\text{engine fuel rate}}{\text{brake power}}$$

$$\text{bsfc (be)} = \frac{\dot{m}_F}{P_b} \left[\frac{\text{g}}{\text{kWh}} \right]$$

$$\text{brake thermal efficiency} = \frac{\text{Energy input}}{\text{Brake power}} = \frac{\text{engine fuel rate} \cdot \text{fuels lower heating value}}{\text{brake power}}$$

$$\text{BTE } (\eta_e) = \frac{\dot{m}_F \cdot LHV}{P_b} \quad \text{Example Diesel Fuel} \quad LHV = 42700 \text{ kJ/kg}$$

$$\text{Excess Air Ratio} = \frac{\text{engine air flow rate}}{\text{engine fuel rate} \cdot L_{min}} \quad \text{Example Diesel Fuel}$$
$$L_{min} = 14,5 \text{ kg}_{\text{Air}}/\text{kg}_{\text{Fuel}}$$

$$\text{EAR } (\lambda) = \frac{\dot{m}_{Air}}{\dot{m}_F \cdot L_{min}}$$

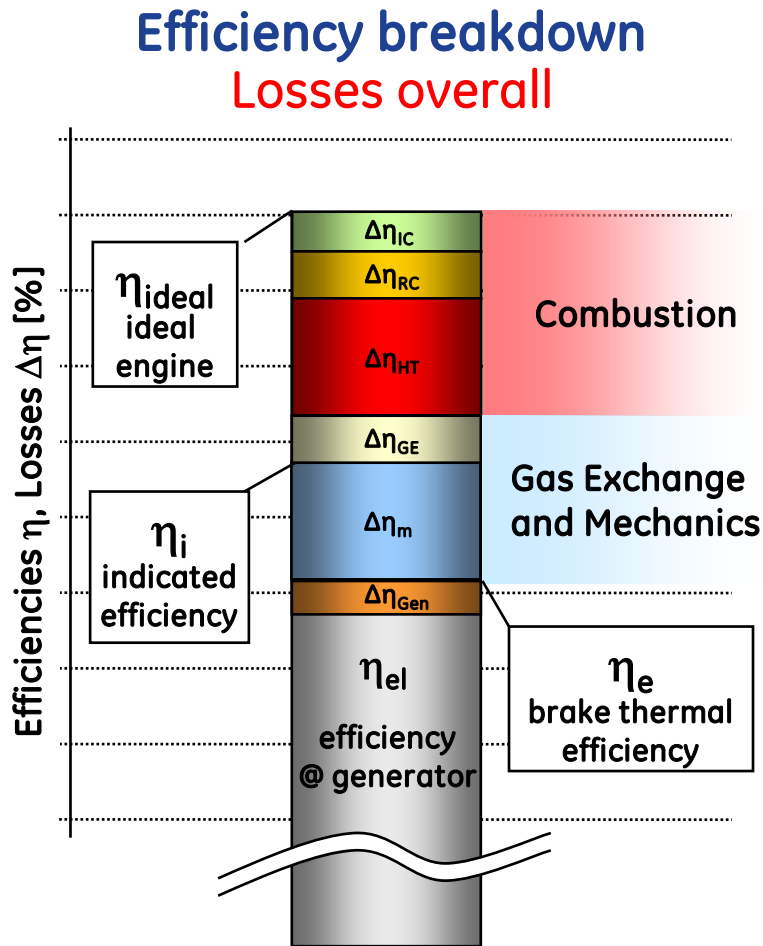


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Performance basics

Real Cycles and losses – Efficiency breakdown



Efficiency losses

$\Delta\eta_{\text{IC}}$	Incomplete combustion
$\Delta\eta_{\text{RC}}$	Real combustion
$\Delta\eta_{\text{HT}}$	Heat losses to cooling sys.
$\Delta\eta_{\text{GE}}$	Gas exchange
$\Delta\eta_m$	Engines friction
$\Delta\eta_{\text{Gen}}$	Generator

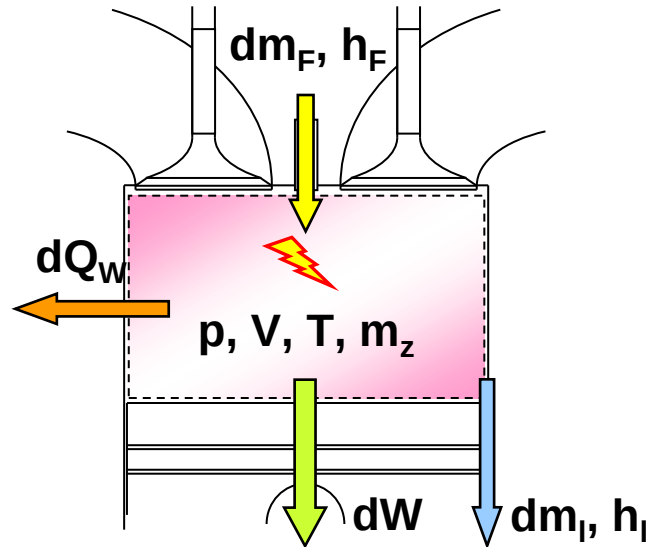


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Performance

Thermodynamics and in cylinder pressure analysis



Measure the in cylinder pressure and calculate...

$$\frac{dQ_B}{dt} = \frac{dU}{dt} - \frac{dQ_W}{dt} + p \frac{dV}{dt} - \frac{dm_l}{dt} h_l - \frac{dm_{F,vapor.}}{dt} \Delta h_{F,vapor.}$$

Heat release

Internal Energy

Wall Heat

Work

Leakage

Fuel vapor



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Performance development

Zieldefinition

Arbeitsschritte:

- Festlegung Motorkonzept (z.B. Gasmotor mager, homogen Lambda=1 Saugrohreinspritzung, geschichtet mager Direkteinspritzung)
- Definition Nennleistung und Nennmoment, Emissionslevel und Verbrauch
- Festlegung relevanter Betriebspunkte für Voll- und Teillast, Definition der Betriebsstrategien für diese Punkte
- Festlegung der Hauptabmessungen (Hub, Bohrung, Verdichtungsverhältniss usw).

Werkzeuge:

- Arbeitsprozessrechnung, Ladungswechselrechnungen
- Datenbanken, Benchmarking, Marktanalysen

Geometriedefinition

Arbeitsschritte:

- Festlegung der Brennraumgeometrie
- Festlegung der notwendigen Massenströme, Auslegung Luftpfad, Aufladesystem, Kanalgeometrien (Füllung + Ladungsbewegung), Steuerzeiten
- Festlegung Kraftstoffeinbringung (Düsen/Mischerauslegung)
- Grundkonstruktion

Werkzeuge:

- 1D-Strömungsberechnung
- 3D-Strömungs- Spray und Verbrennungsrechnung
- Design of Experiments
- CAD

Konzeptevaluation

Arbeitsschritte:

- Evaluation des Konzepts am Motor: Überprüfung von Füllung, Ladungsbewegung, Kraftstoffeinbringung und Verbrennung durch experimentelle Untersuchungen
- Detailkonstruktion

Werkzeuge:

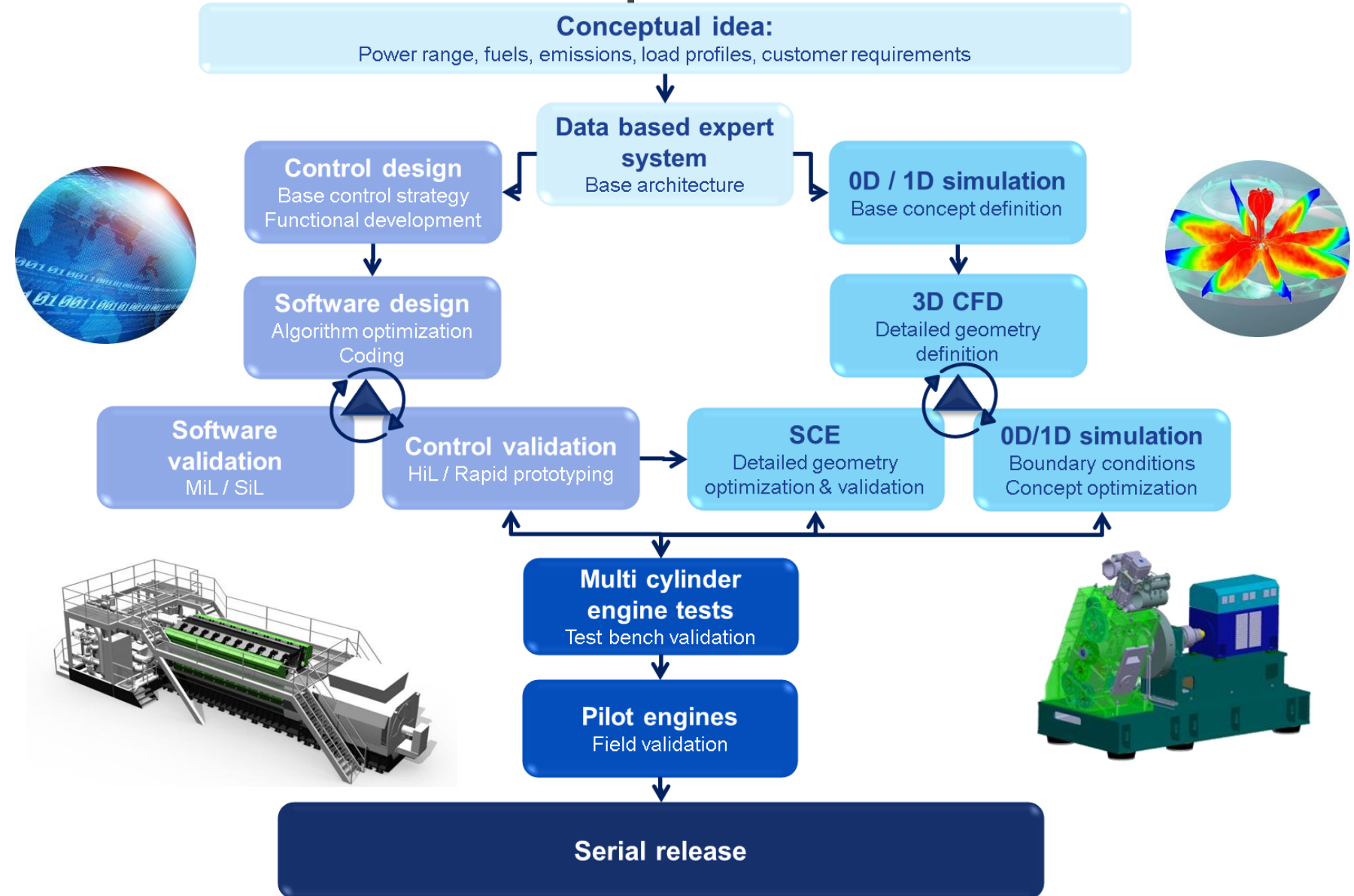
- Druckkammern
- Optischer Einzylindermotor
- Thermodynamischer Einzylindermotor
- Vollmotor
- PIV, LIEF, Schlierendiagnostik
- Endoskopie / Videoskopie
- Indiziermesstechnik
- Thermodynamische Analysen
- Design of Experiments



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Performance development chain



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Model Classification

- 1) Best estimate/approximation
- 2) Transfer function
- 3) Lumped parameter model
- 4) 0D Phenomenological model
- 5) 1D Phenomenological model
- 6) 3D CFD + simple combustion
- 7) 3D CFD + complex chemistry

Classification of Engine Models

Spatial classification

Multi-dimensional

3-D CFD
(e.g. CONVERGE, Fluent)



One-dimensional

1-D CFD (e.g. GT-Power)

**Multi-zone
combustion**

Zero-dimensional

Lumped volume models
Uniform cylinder charge

**Single-zone
combustion**

Decreasing detail, faster execution, more control-oriented

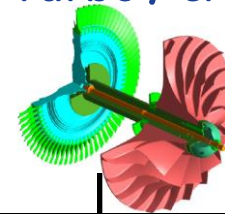
Combustion



Manifold pressure



Turbo / crankshaft
dynamics



Temporal
classification

10^{-3}

10^{-2}

10^{-1}

Time scale (seconds)

10^0

10^1

Sub-Cycle

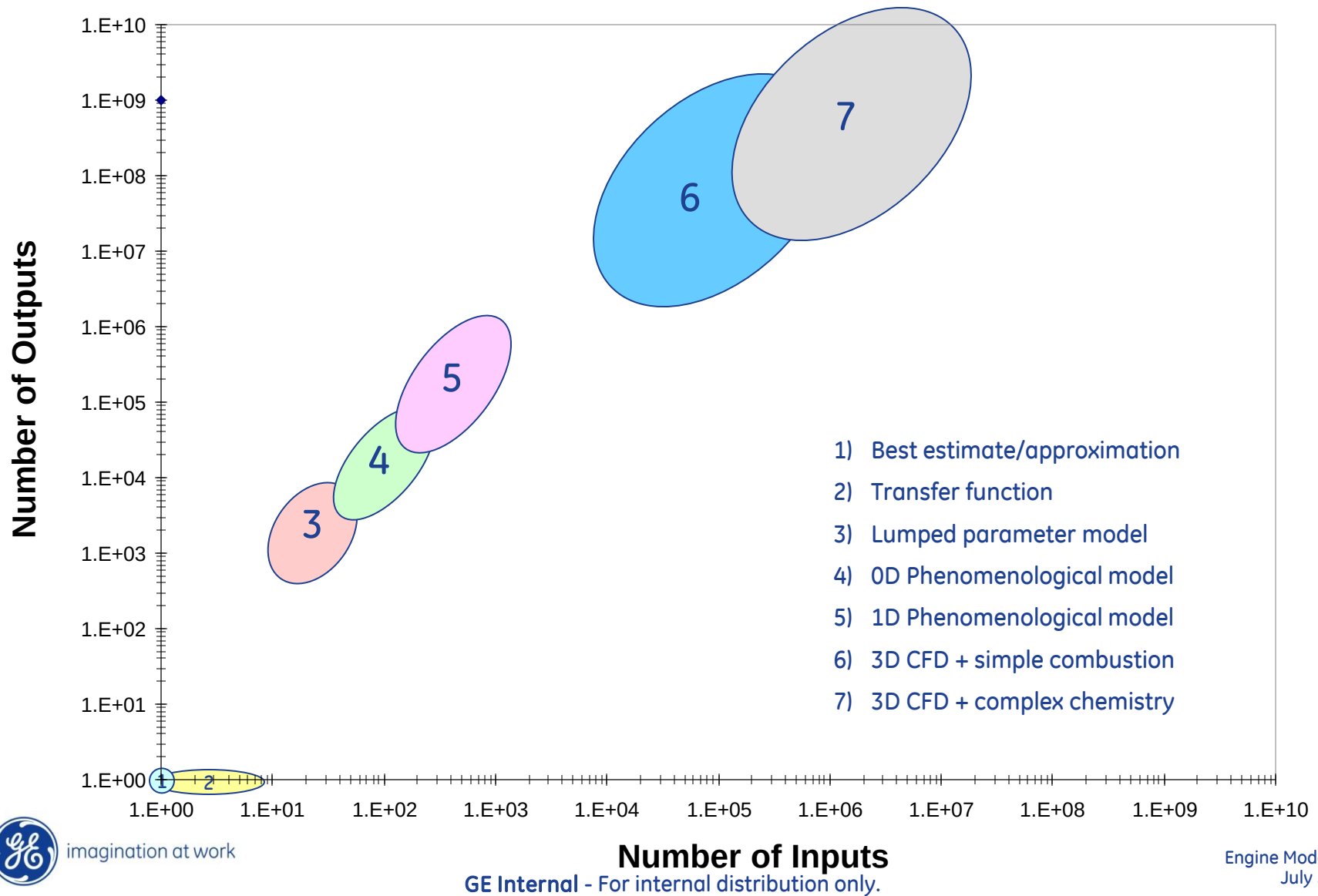
Cycle

Many Cycles

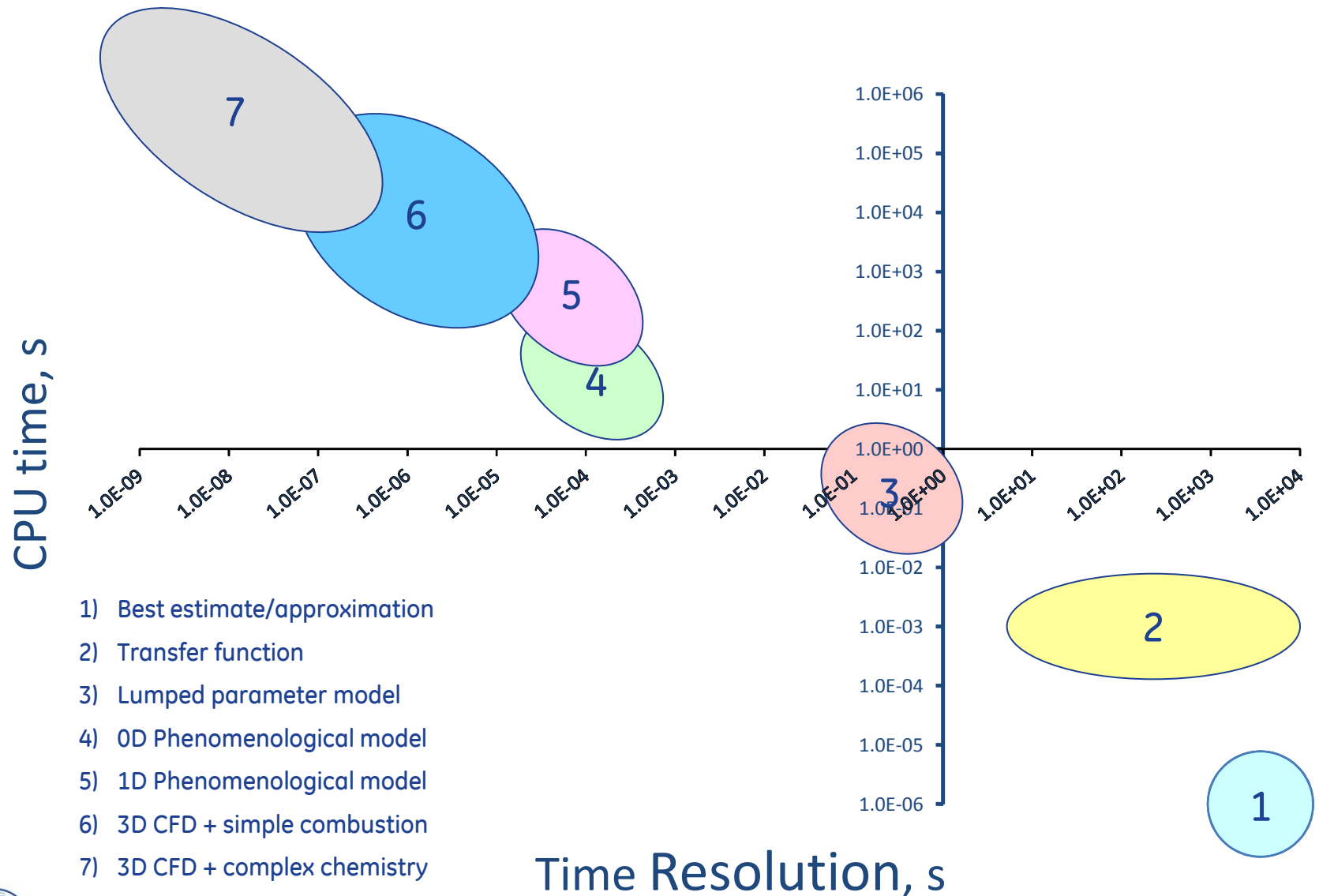
Model Classification

	Model Type	Number of inputs	Physical description	Operating conditions	Model Boundaries	Model Resolution	Outputs	Time Resolution	CPU Requirement	Computer Memory Required
1	Best estimate (approximation)	0	none	considered	complete system	complete system	fixed values	n/a	0	0
2	Transfer function	1-7	buried in transfer function	considered	complete system or sub-system	complete system or sub-system	single value per function	n/a	0.001 seconds	kB
3	Lumped parameter	10-50	minimal	minimal	complete system	individual component	time resolved component level values	1.00E-01	0.1-1.0 seconds	kB
4	0D phenomenological (filling & emptying)	50-200	moderate	moderate	complete system	individual component	crank angle resolved values within some components	1.00E-04	0.2-20 seconds	kB
5	1D phenomenological (1D wave dynamics)	200-1000	detailed	moderate	complete system	sub-component	sub-crank angle resolved values within all components, 1D spatially resolved values for ducting	5.00E-05	2-20 minutes	MB
6	3D Fluid Mechanics + simple combustion	10,000+	complex	complex	component	sub-component	micro-crank angle and 3D spatially resolved values for all complete domain	1.00E-06	2-14 hours	GB
7	3D Fluid Mechanics + complex chemistry	10,000+	complex	complex	component	sub-component	micro-crank angle and 3D spatially resolved values for all complete domain	1.00E-09	2-14 days	GB

Model Data Intensity



Resolution and CPU Time



Best estimate/approximation

- Draw on past experience
- Interpretation of available data
- Typically set for behavior of a complete system
- Produces a single fixed (time-invariant) value for particular set of circumstances
- "Processing" is handled by the individual estimating

Transfer function

- Formulated from data sets (test, analysis)
- Inputs are typically selected operating conditions or design parameters
- Typically a complete system but may be set up for looking at subsystems
- Good for interpolation within experience (should not use to extrapolate)
- A single, time-invariant, value is determined for each function
- Very little processing time required (milliseconds)
- Very low computer storage required (kB)

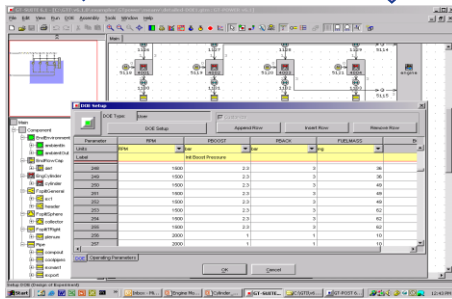
Lumped parameter (mean value) model

- A simplified time-variant modeling approach
- Used in control system logic
- Minimal description of the physical system is required
- Conducted for complete system response
- Model resolved to an individual component level
- Time resolved values at component connections available as output
- Time resolution typically on the order of 1 second
- Very low CPU time required (less than 1 second)
- Very low computer storage required (kB)

Control Oriented Mean Value Engine Model (MVEM)

DOE Set-up

Combustion Model



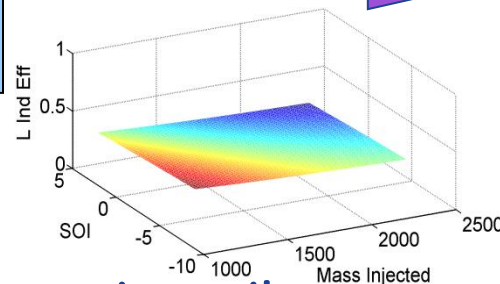
GT-Power Model (1-D CFD)

Train/build static regression maps

Physics-based *Interaction models* (turbo, crankshaft, manifold)

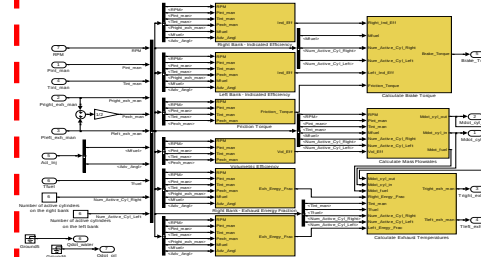
Combustion maps

- Volumetric efficiency
- Thermal efficiency
- Exhaust energy fraction
- Friction



Model Reduction Process

Modular Simulink-Based Control-Oriented Models



Field data

Emissions Model

Common Rail Fuel System Model

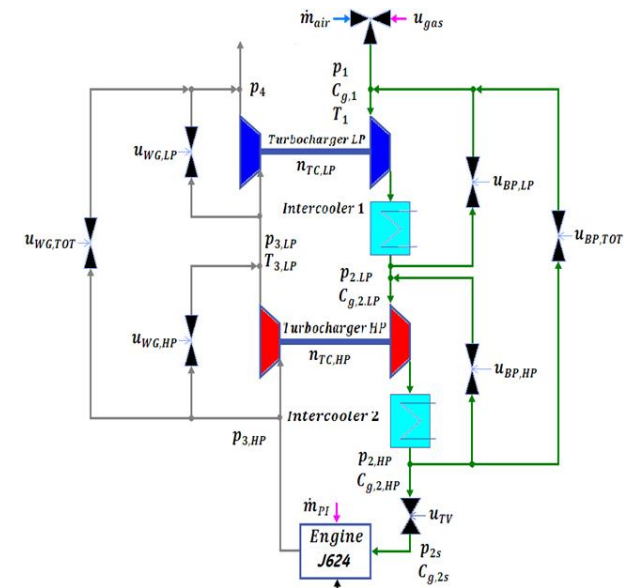
Process is easily repeatable for new engines

Tier 2 in 2009; Tier 3 in 2010 & Tier 4 in 2011

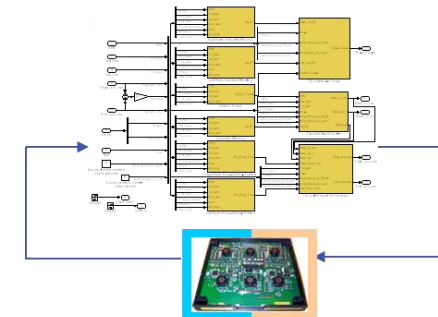
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Applications of MVEM

- ECU s/w Validation with Hardware in the Loop (HiTL)
- Sensors & Actuators selection
- Control design tools
- Model-in-Loop (MiL) studies
- Model Based System Design
- ECU models for diagnostics



System Architecture

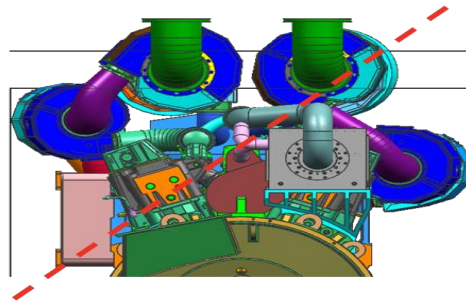
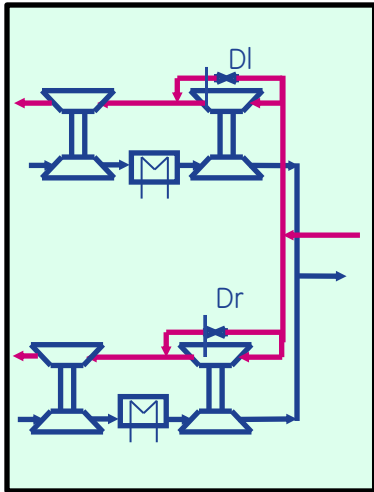


ECU

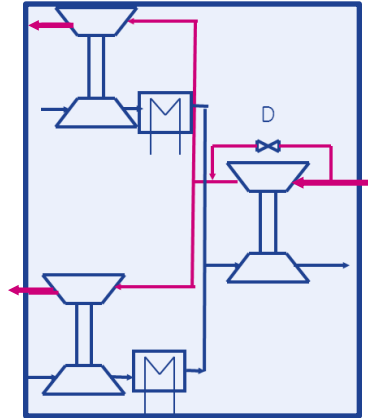
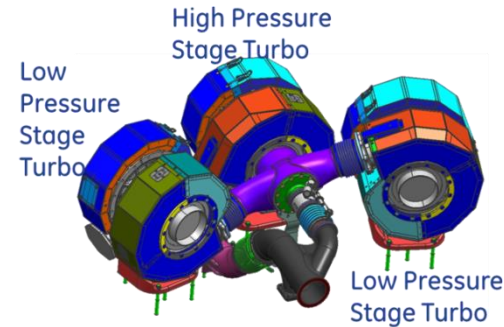
Enable Control design, capture system interactions early

Model Based Design: Marine Tier 4 Example

Four Turbo Configuration



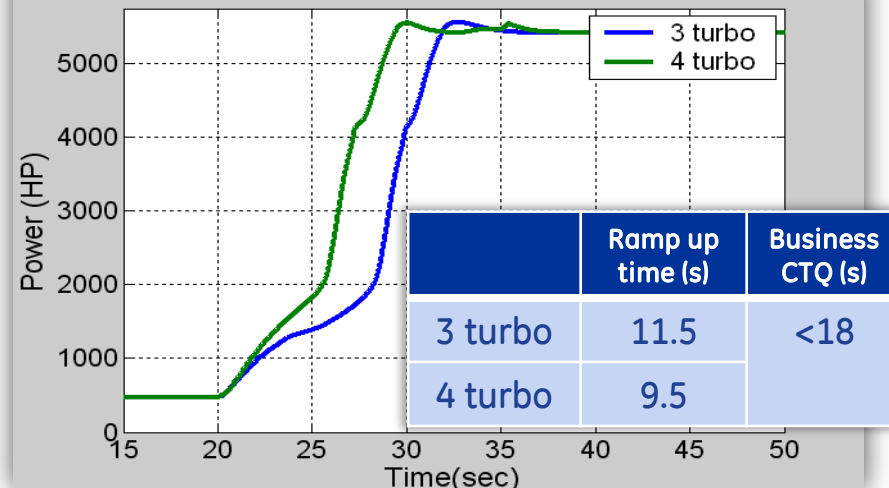
Three Turbo Configuration



Results from Simulink Mean Value Model

Trade off matrix	Three turbo	Four turbo
Cost, packaging & system complexity	Lower	Higher
Transient response	Slower but well within CTQ	Faster & well within CTQ

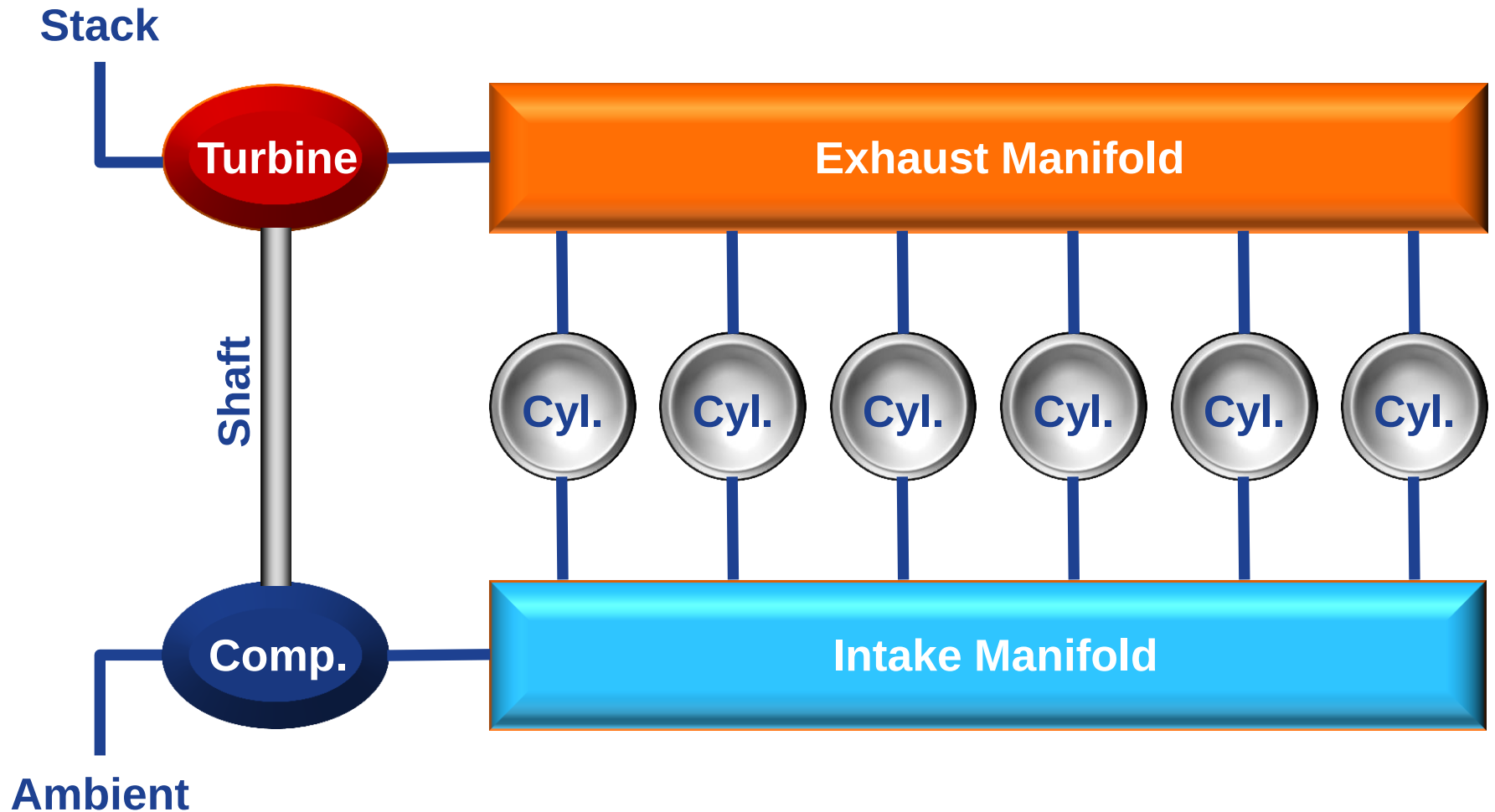
Model based design use for turbo layout selection



0D Phenomenological model

- Often referred to "filling & emptying"
- System broken into discrete components with mass & energy transfer connections
- "Active" components connected via control volumes (manifolds)
- Each active component modeled with sufficient detail to capture main behavior
- Moderate level of physical description required for each component
- Conducted for complete system response (boundary can be drawn for subsets of model)
- Model resolved to an individual component level
- Time resolved values for each component available as output
- Time resolution typically on the order of 1 crank degree
- Low CPU time required (less than 1 minute)
- Very low computer storage required (kB)

0D Phenomenological model



1D Phenomenological model

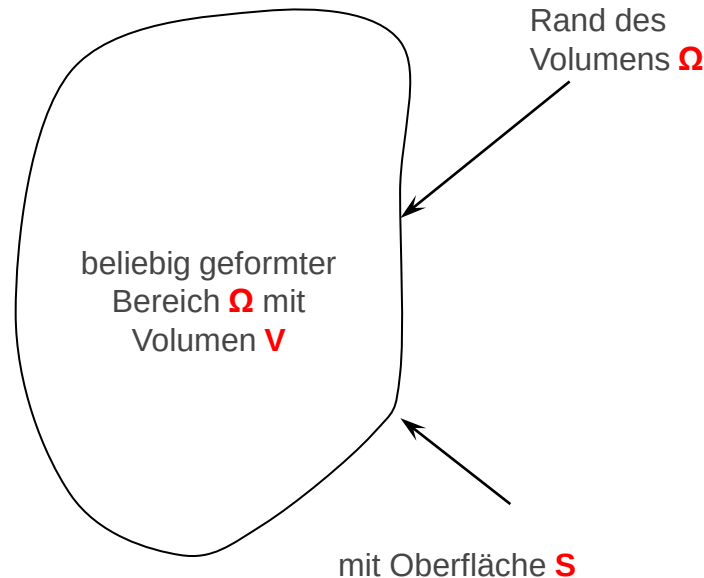
- Often referred to as "wave dynamics modeling"
- Similar to 0D model except spatial resolution added to connections between components
- System broken into discrete components with mass & energy transfer connections
- "Active" components connected via pipes with 1D wave dynamics
- Each active component modeled with sufficient detail to capture main behavior
- Detailed physical description required for each component
- Conducted for complete system response (boundary can be drawn for subsets of model)
- Model resolved to individual components and spatially within ducting
- Time resolved values for each component and piping grid point available as output
- Time resolution typically on the order of fractions of crank degrees
- Moderate CPU time required (30 minutes)
- Moderate computer storage required (MB)

Navier-Stokes-Equations

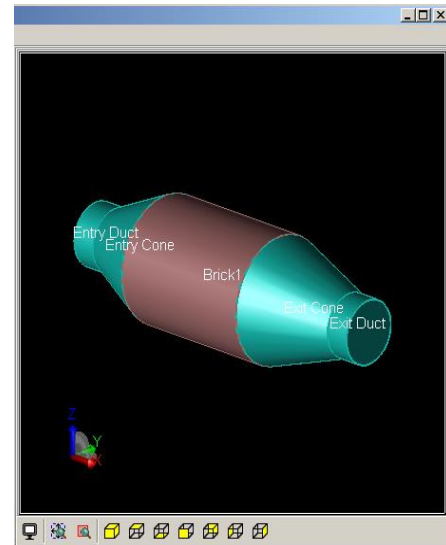
Allgemeine Erhaltungsgleichung:

Voraussetzung ist, dass ein gasgefülltes Volumen als Kontinuum betrachtet wird, um dadurch den Umgang mit infinitesimalen Größen zu ermöglichen.

Zu jedem Zeitpunkt t lässt sich eine physikalische Größe $F(t)$ für Ω angeben (z.B. Gesamtmasse, Gesamtimpuls).



Quelle: Ricardo WAVE



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Navier-Stokes-Equations

Die physikalische Größe $F(t)$ kann auch beschrieben werden, indem man ihre zugehörige Dichte $f(r,t)$ über Ω integriert.

Mit Dichte ist nicht zwingend die Massendichte gemeint, sondern die Größe $F(t)$ pro Volumen:

$$F_{(t)} = \int_{\Omega} f(r,t) dV \quad r = \text{Ortsvektor}$$

Die Änderung der Größe $F(t)$ mit der Zeit wird durch folgenden Ausdruck beschrieben:

$$\frac{\partial F_{(t)}}{\partial t} = \int_{\Omega} \frac{\partial f_{(r,t)}}{\partial t} dV$$



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Navier-Stokes-Equations

Die Änderung der Größe $F(t)$ mit der Zeit kann erfolgen durch:

1. Die Größe $F(t)$ strömt durch die Oberfläche in das Volumen hinein oder aus dem Volumen heraus. Die Strömung selbst wird dabei durch den Stromdichtevektor Φ_f beschrieben. Veränderung der Dichte ist gegeben durch $-\Phi_f \cdot ndS$
2. Im Inneren des Gebiets Ω liegt eine Quelle bzw. Senke von $F(t)$. So wird z.B. eine chemische Reaktion beschrieben. Dabei beschreibt ein Quellenterm q_f die pro Zeit t und Volumeneinheit gebildete Menge an F .
3. Fernwirkung beeinflusst das Innere von Ω z.B. durch Gravitation oder Wärmestrahlung. Dabei beschreibt ein Fernwirkungsterm s_f die pro Zeit t und Volumeneinheit gebildete Menge an F .



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Navier-Stokes-Equations

Die Änderung von $F(t)$ für den jeweiligen Prozess ergibt sich durch die Integration über die Oberfläche von $\partial\Omega$ bzw. über Ω selbst.

Die gesamte Änderung von $F(t)$ pro Zeit t ergibt sich durch Aufsummieren der Einzelintegrale:

$$\frac{\partial F_{(t)}}{\partial t} = \int_{\Omega} \frac{\partial f_{(r,t)}}{\partial t} dV = - \int_{\partial\Omega} \Phi_f \cdot n dS - \int_{\Omega} q_f dV + \int_{\Omega} s_f dV$$

Zeitliche Änderung Änderung durch Strömen Änderung durch Quellen / Senken Änderung durch Fernwirkung



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Navier-Stokes-Equations

Durch Vereinfachung erhält man die allgemeine Erhaltungsgleichung:

$$\frac{\partial f}{\partial t} + \operatorname{div} \Phi_f = q_f + s_f$$

Im Folgenden wird allgemeine Erhaltungsgleichung für Masse, Impuls und Energie anstelle von $F(t)$ angewandt.



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Navier-Stokes-Equations

Erhaltung der Gesamtmasse:

$$\frac{\partial p}{\partial t} + \operatorname{div}(pv) = 0$$

pv = Massenstromdichte

Erhaltung des Gesamtimpulses:

$$\frac{\partial(pv)}{\partial t} + \operatorname{div}(pv \otimes v) + \operatorname{div} p^{\bar{}} = pg$$

Erhaltung der Gesamtenergie:

$$\frac{\partial(pe)}{\partial t} + \operatorname{div}(pev + p^{\bar{}} - \lambda \operatorname{grad} T) = q_r$$



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Navier-Stokes-Equations

Die drei aufgestellten Differentialgleichungen zum Berechnen von Strömungen (Massen-, Impuls- und Energieerhaltungsgleichungen) lassen sich wie folgt in einem Gleichungssystem zusammenfassen:

$$\frac{\partial}{\partial t} \begin{pmatrix} p \\ pv \\ pe \end{pmatrix} + \operatorname{div} \begin{pmatrix} pv \\ p \otimes v + p^= \\ pev + p^= - \lambda \operatorname{grad} T \end{pmatrix} = \begin{pmatrix} 0 \\ pg \\ q_r \end{pmatrix}$$

Dieses Gleichungssystem bildet die Grundlage für die ein- oder mehrdimensionale Simulation von Strömungen.



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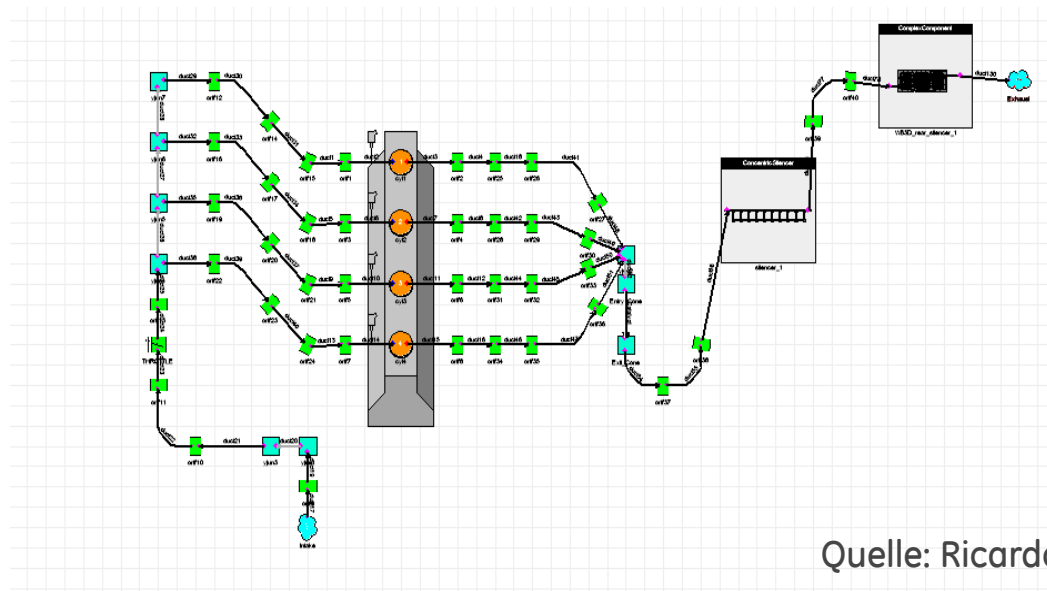
1D simulation

Die Gesamtheit der kontinuumsmechanischen Erhaltungssätze für

- Masse
- Impuls und
- Energie

in Strömungsfeldern werden als **Navier-Stokes-Gleichungen** bezeichnet.

Sie bilden die Grundlage für die Simulation strömungsmechanischer Vorgänge.



Quelle: Ricardo WAVE



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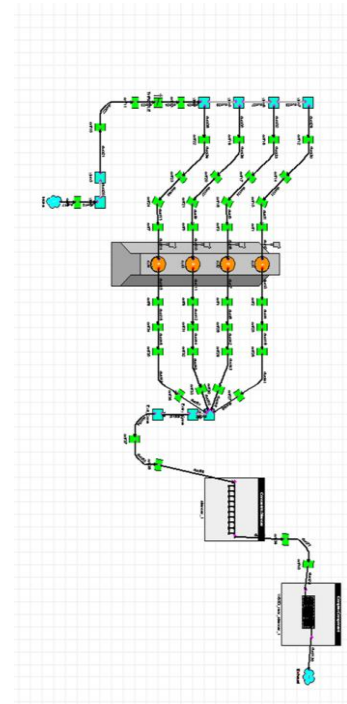
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1D simulation

Tools zu eindimensionalen Strömungssimulation

eindimensional: **Navier-Stokes-Gleichungen** werden für Größen gelöst, die quer zur Hauptströmungsrichtung gemittelt sind.

Vorteil: Die eindimensionale Abbildung strömungsrelevanter Bauteile in Verbindung mit nulldimensionalen Verbrennungsmodellen ermöglicht z.B. die Berechnung ganzer Kennfelder mit vertretbarem zeitlichem Aufwand.



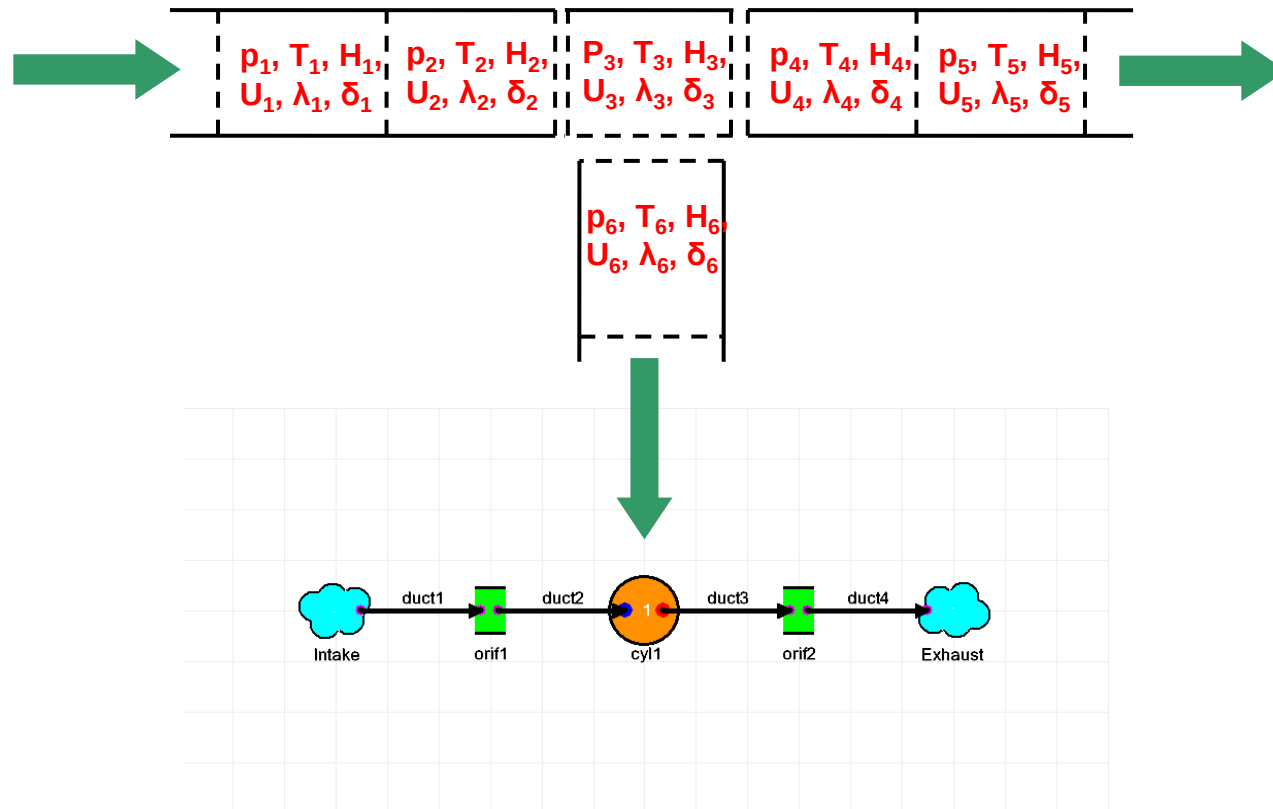
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1D simulation

Eindimensional bedeutet, dass die Navier-Stokes-Gleichungen für Größen gelöst werden, die über die Strömungsrichtung gemittelt sind.

Dazu wird das zu simulierende System diskretisiert:



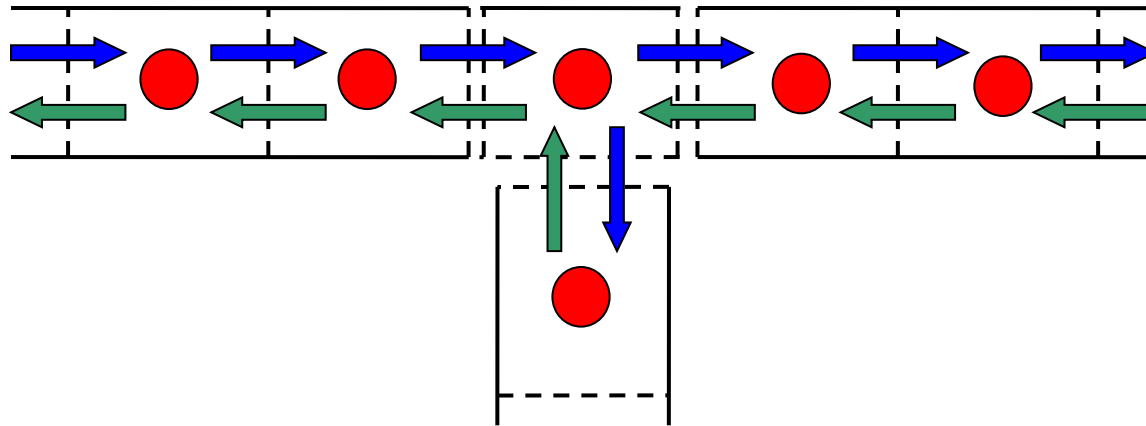
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1D simulation

Es wird angenommen, dass die Variablen über das diskretisierte Volumen gleich sind. Berechnet werden die Variablen für den Volumenmittelpunkt.

Die einzelnen Volumen sind im Rechenmodell so miteinander verknüpft, dass eine Interaktion der Variablen möglich ist.



Während der Simulation wird dann so lange gerechnet bis definierte Größen konvergieren.



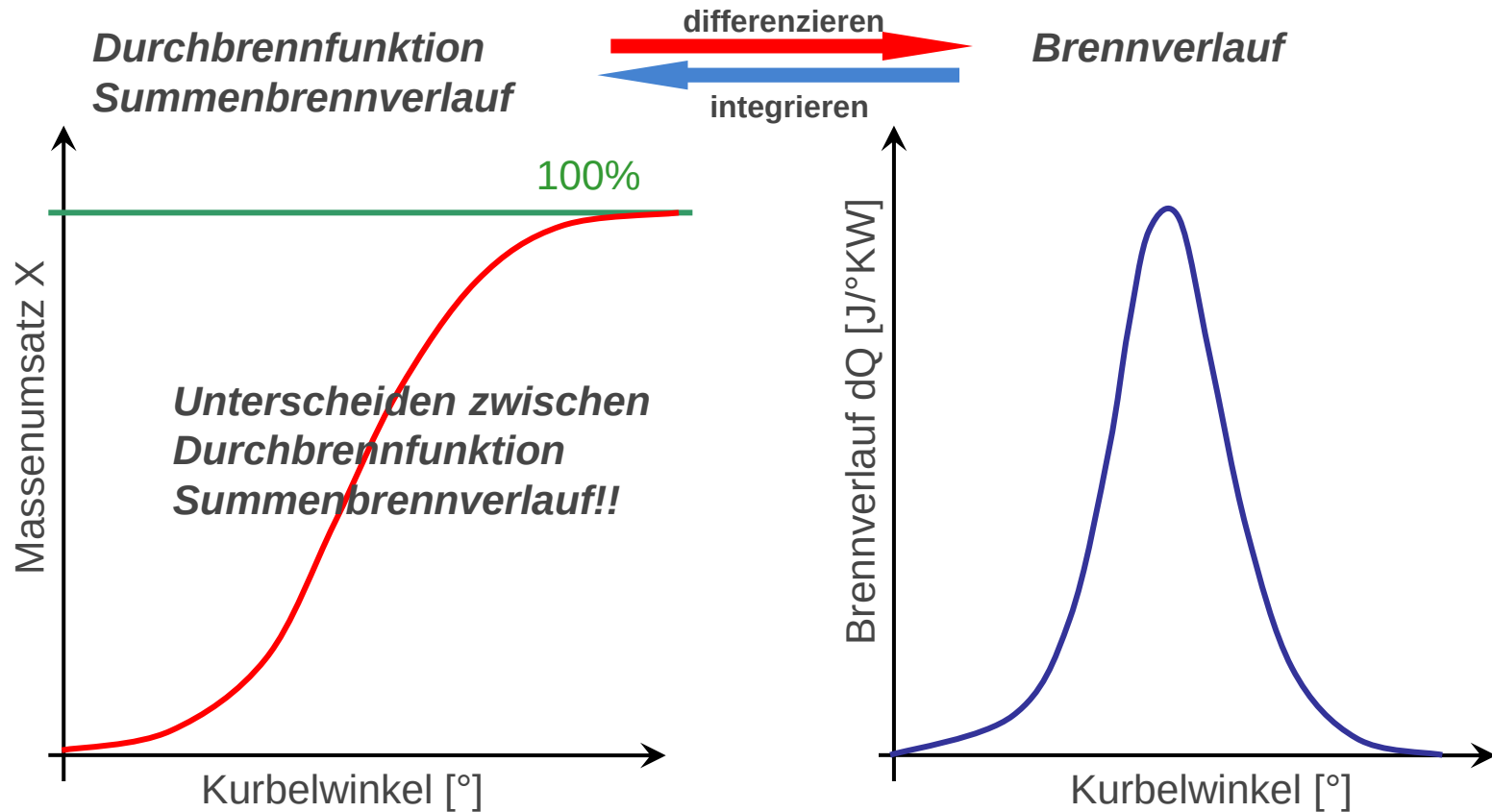
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Quelle: Ricardo WAVE

1D simulation : Combustion estimations

Brennverlauf beschreibt zeitlichen Verlauf der Energiefreisetzung



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1D simulation : Combustion estimations

In der Vergangenheit gab es zwei Methoden, mit denen versucht wurde, den Brennverlauf zu berechnen:

Approximation der
Verbrennungslinie durch
beliebige mathematische
Gleichungen



Aussage über Einfluss
einzelner Einflussfaktoren
nicht möglich

Berechnung der
Verbrennungslinie anhand von
Gleichungen der Thermodynamik
und der Energiefreisetzungs-
geschwindigkeit



Entsprechende
Aussage ist prinzipiell
möglich



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1D simulation : Combustion estimations

Die Erstellung eines Brennverlaufmodells erfordert die Kenntnis der Verbrennungsgeschwindigkeit.

Ansätze:

-Empirische Gleichungen von Neumann und Goncar

-Bimolekulare Gleichungen von Neumann und Kuschul, weiterentwickelt von Inozemcev



-Darstellung der Verbrennung als Kettenreaktion durch

→ autogenetische Gleichungen nach Akulov

→ autokatalytische Gleichungen nach Semenov und Koskin



← **freie Radikale**



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1D simulation : Combustion estimations

Vibe entwickelt auf Grundlage der Annahme der Verbrennung als Kettenreaktion eine eigene Gleichung der **Verbrennungsgeschwindigkeit**:

$$w = k \cdot n \cdot t^m \cdot e^{-\frac{k \cdot n}{m+1} t^{m+1}}$$

Es bedeuten :

w = *Verbrennungsgeschwindigkeit*

k = *Proportionalitätsfaktor*

n = *Drehzahl*

m = *Kennwert des Verbrennungscharakters*



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1D simulation : Combustion estimations

Summenbrennverlauf:

Im nächsten Schritt formuliert Vibe die Energiefreisetzung über °KW in konventionellen Otto- und Dieselmotoren als Summenbrennverlauf:

$$\frac{Q_B(\varphi)}{Q_{B,ges}} = 1 - e^{-a \left(\frac{\varphi - \varphi_{bb}}{\Delta\varphi_{BD}} \right)^{m+1}}$$

Es bedeuten :

$Q_B(\varphi)$ = bis zum Zeitpunkt φ umgesetzte Wärmemenge

$Q_{B,ges}$ = gesamte während der Verbrennung freigesetzte Energie

φ_{bb} = Brennbeginn

$\Delta\varphi_{BD}$ = Brenndauer

a = Anpassungsfaktor $\Rightarrow a = \ln(1 - \eta_{ges})$

η_{ges} = Umsetzungsgrad

m = Kennwert des Verbrennungscharakters



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1D simulation : Combustion estimations

Brennverlauf:

– Wird der Summenbrennverlauf nach °KW differenziert, erhält man den Brennverlauf:

$$\frac{dQ_B}{d\varphi} = Q_{B,ges} \cdot a \cdot (m+1) \cdot \left(\frac{\varphi - \varphi_{BB}}{\Delta\varphi_{BD}} \right) \cdot e^{-a \left(\frac{\varphi - \varphi_{bb}}{\Delta\varphi_{BD}} \right)^{m+1}}$$

Die Anpassung der Vibe-Parameter erfolgt so, dass wichtige Prozessgrößen (p_{max} , p_{mi} , T_{Abgas}) in Übereinstimmung mit der Realität abgebildet werden.

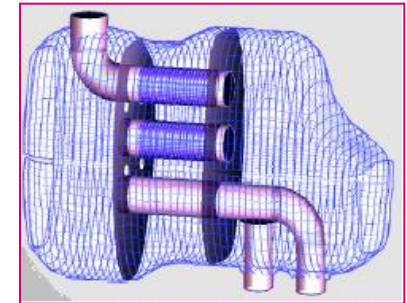


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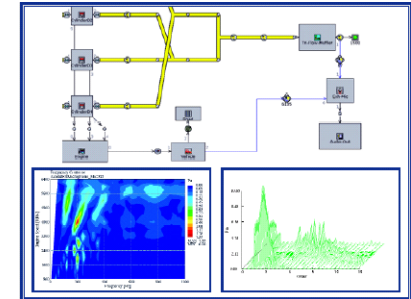
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GT-Power

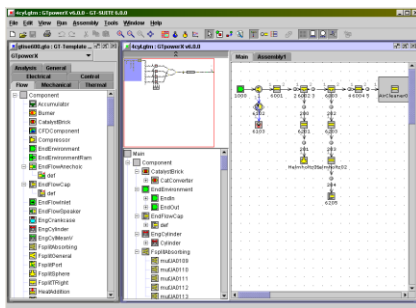
- Network of thermodynamic elements
- Mass, momentum and energy transport via 0/1D connections
- 0, 1D or 3D treatment for major components



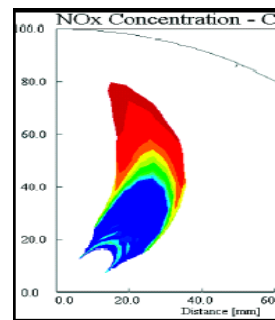
Manifold design



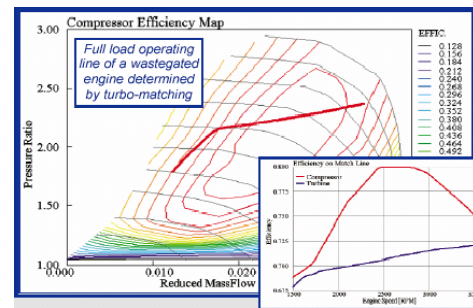
Acoustics



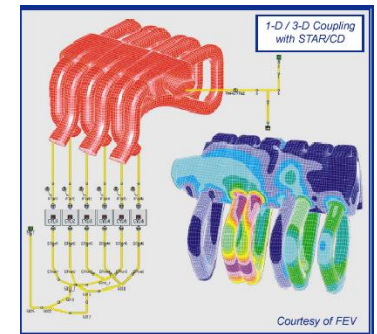
Engine analysis



Emission

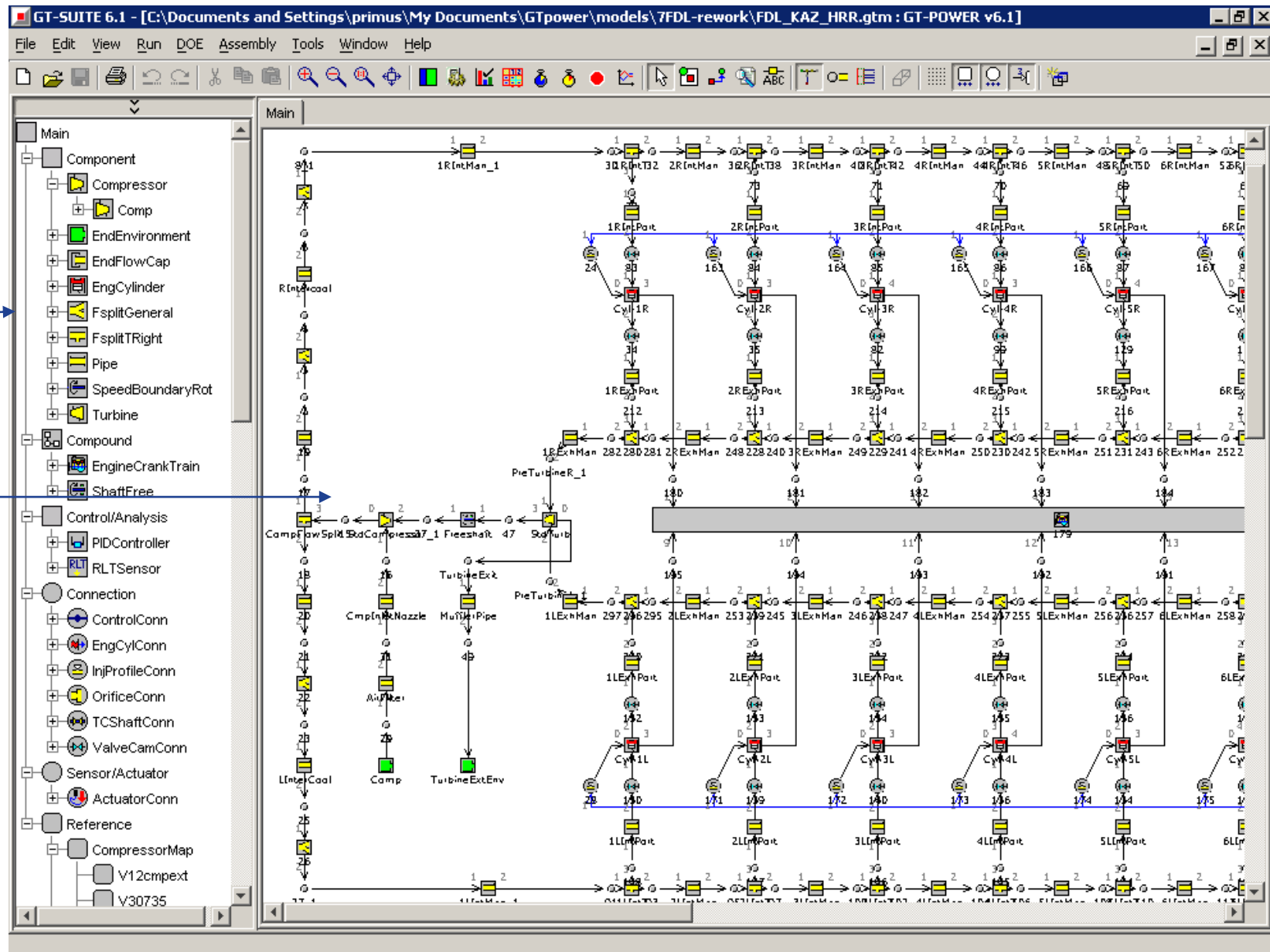


Turbocharger matching

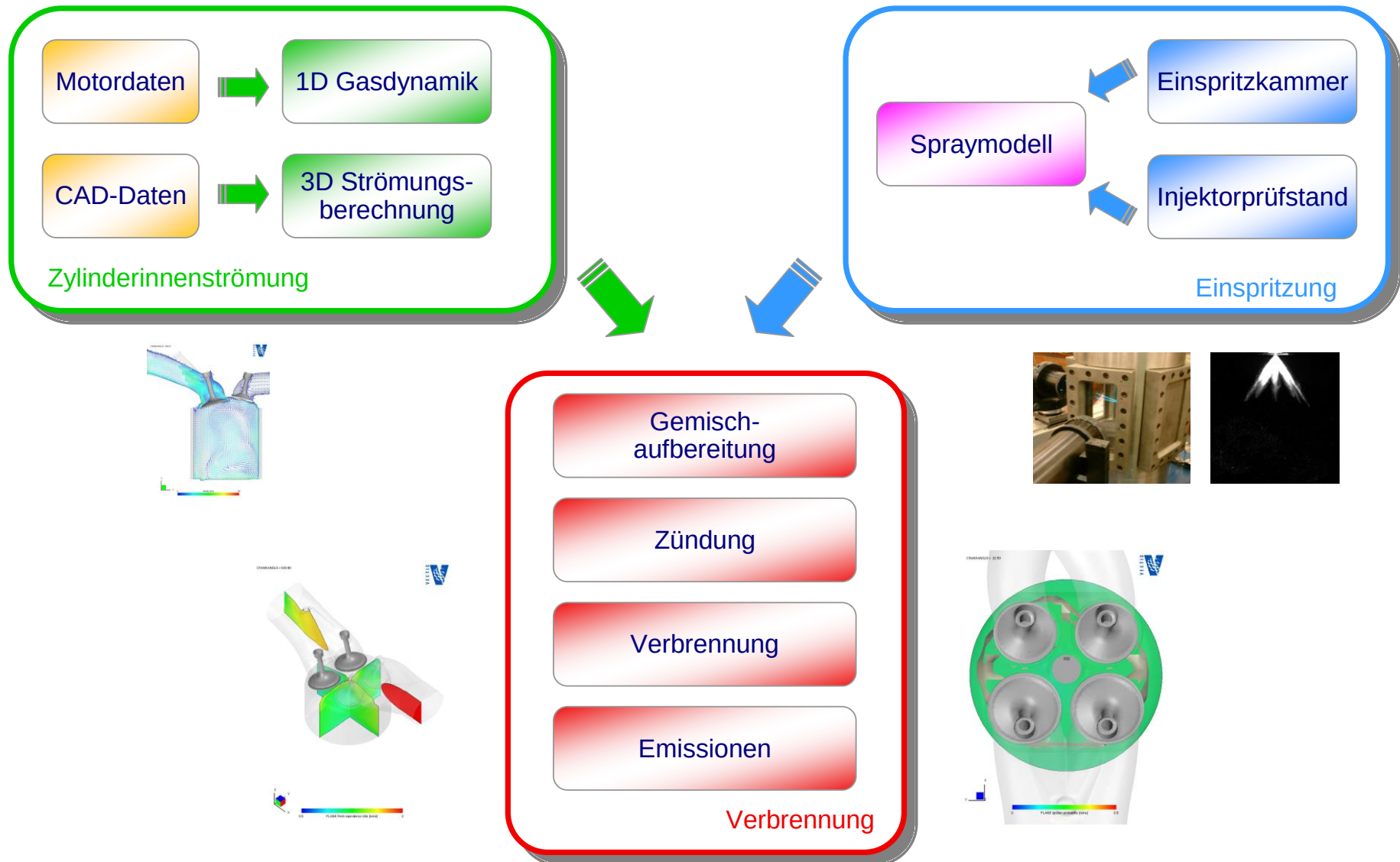


1D/3D integration

GT-Power Model



Performance simulation tool chain

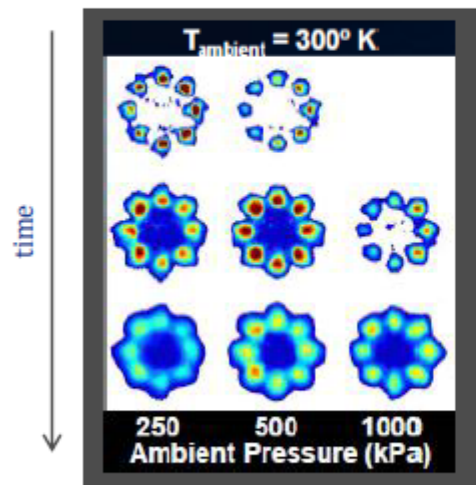


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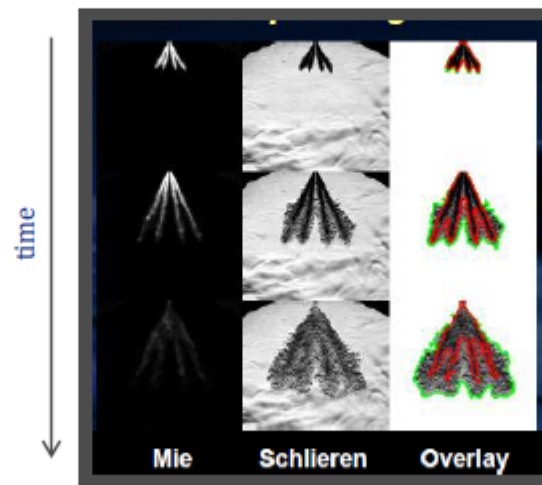
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Spray model development

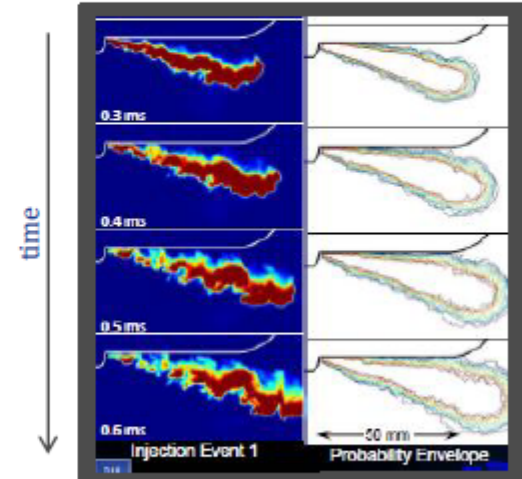
- High temperature pressure vessel
[0.2 – 100 Bar; 300 - 1000 K; continuously purged (0%-21% O₂)]
- Advanced diagnostics for spray characterization



Planar Extinction
Tomography



Near-Simultaneous MIE-
Schlieren Imaging



Near-Simultaneous MIE-
PLIF Imaging



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See tutorial regarding

3D CFD + simple reactions

- Typically conducted for a sector of a given cylinder
- A 3D grid used to section the gaseous/fluid mechanics area of the cylinder
- Grids typically between 100,000 and 1,000,000 cells
- Detailed cylinder geometry, fuel injection, and boundary conditions required
- Individual cell data temporally and spatially resolved available as output
- Time resolution typically on the order of 10 E-6 seconds
- High CPU time required (2-140 hours)
- High computer storage required (GB)

3D Computational fluid dynamics

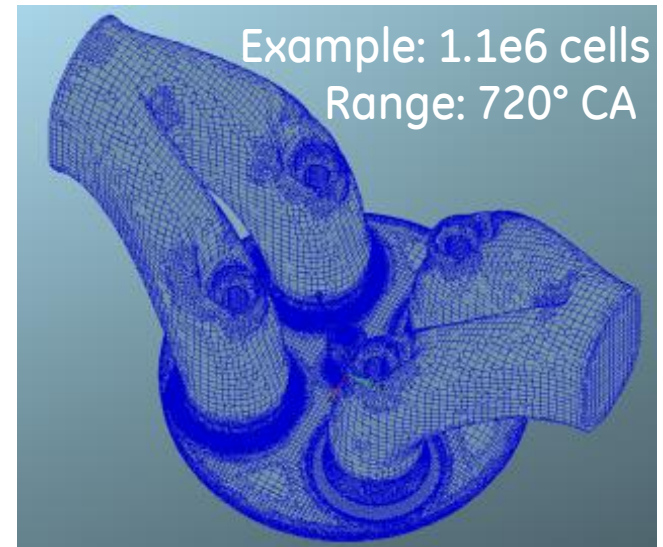
Comparison of different architectures

12/2012 state-of-the-art 12 CORE

Lec_171, Intel Xeon X5690@3.47 Ghz: **63.6h**

02/2014 state-of-the-art 16 CORE

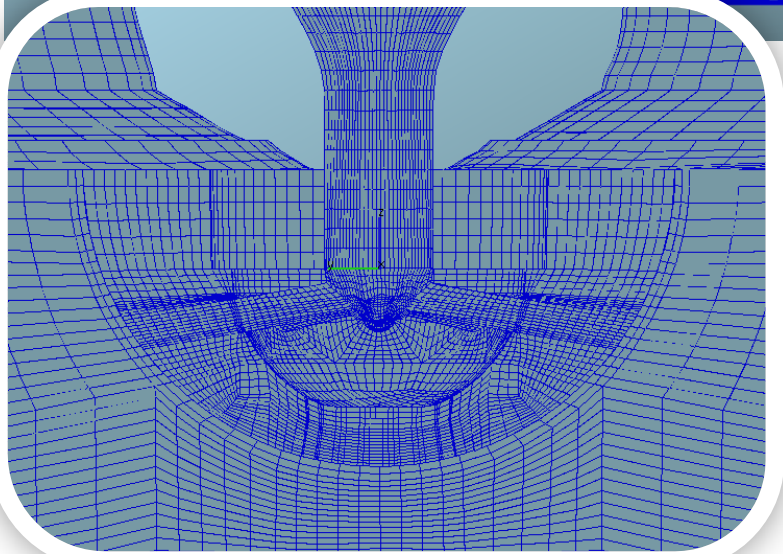
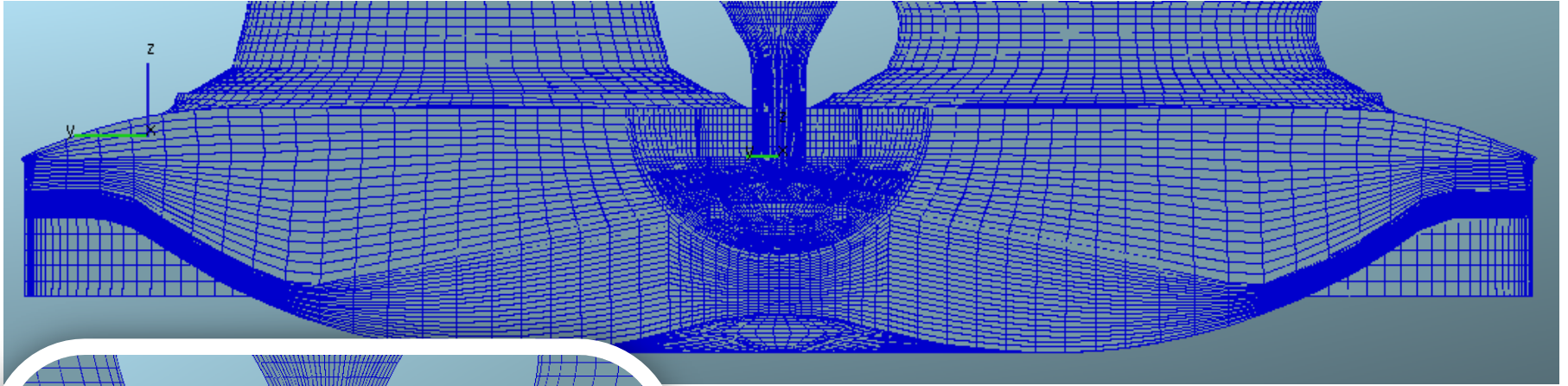
Lec_182, Intel Xeon E5-2687W@3.10 Ghz: **43.8h**



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Structured Mesh I

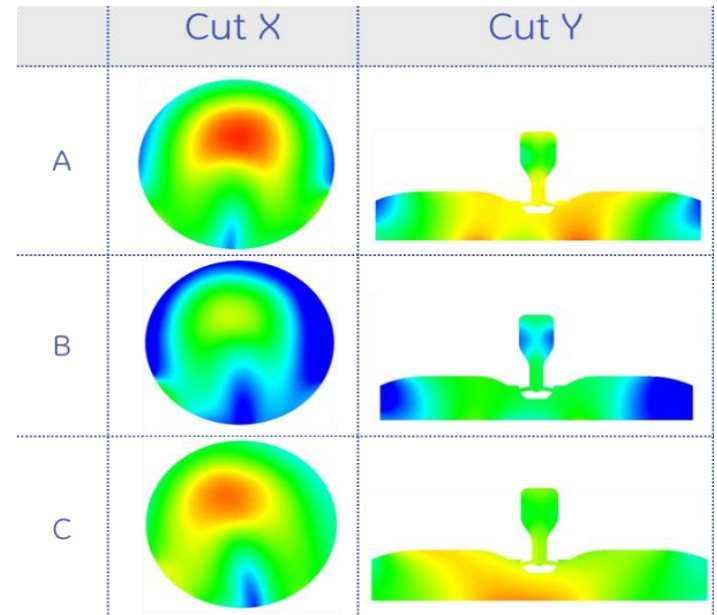
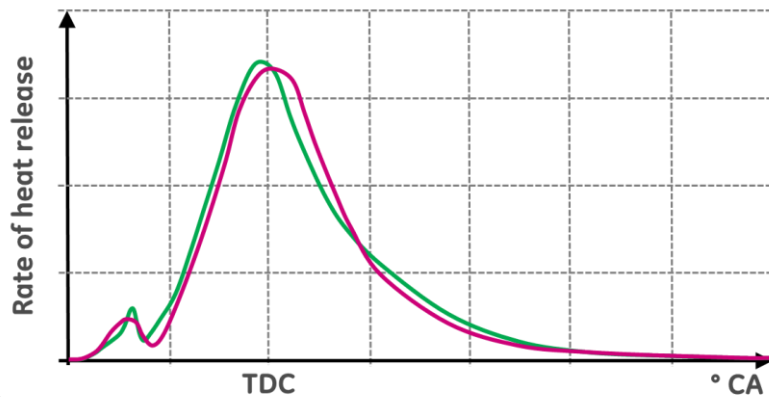
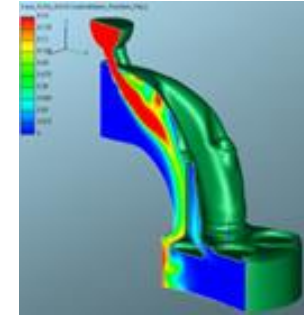
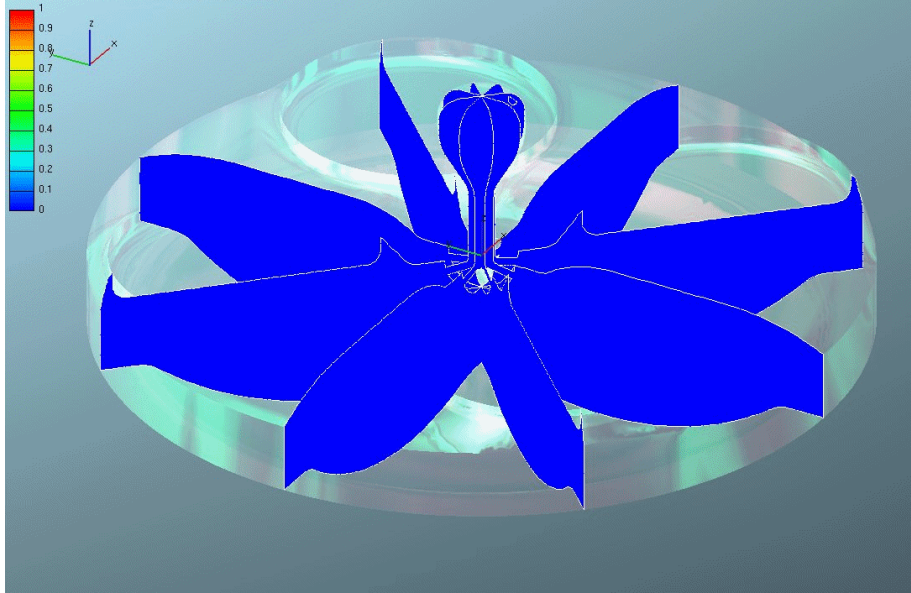


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3D Computational fluid dynamics

Case_11-AN_696.0 Comb-Reaction_Progress_Variable[-]



In cylinder equivalence ratio distribution at firing



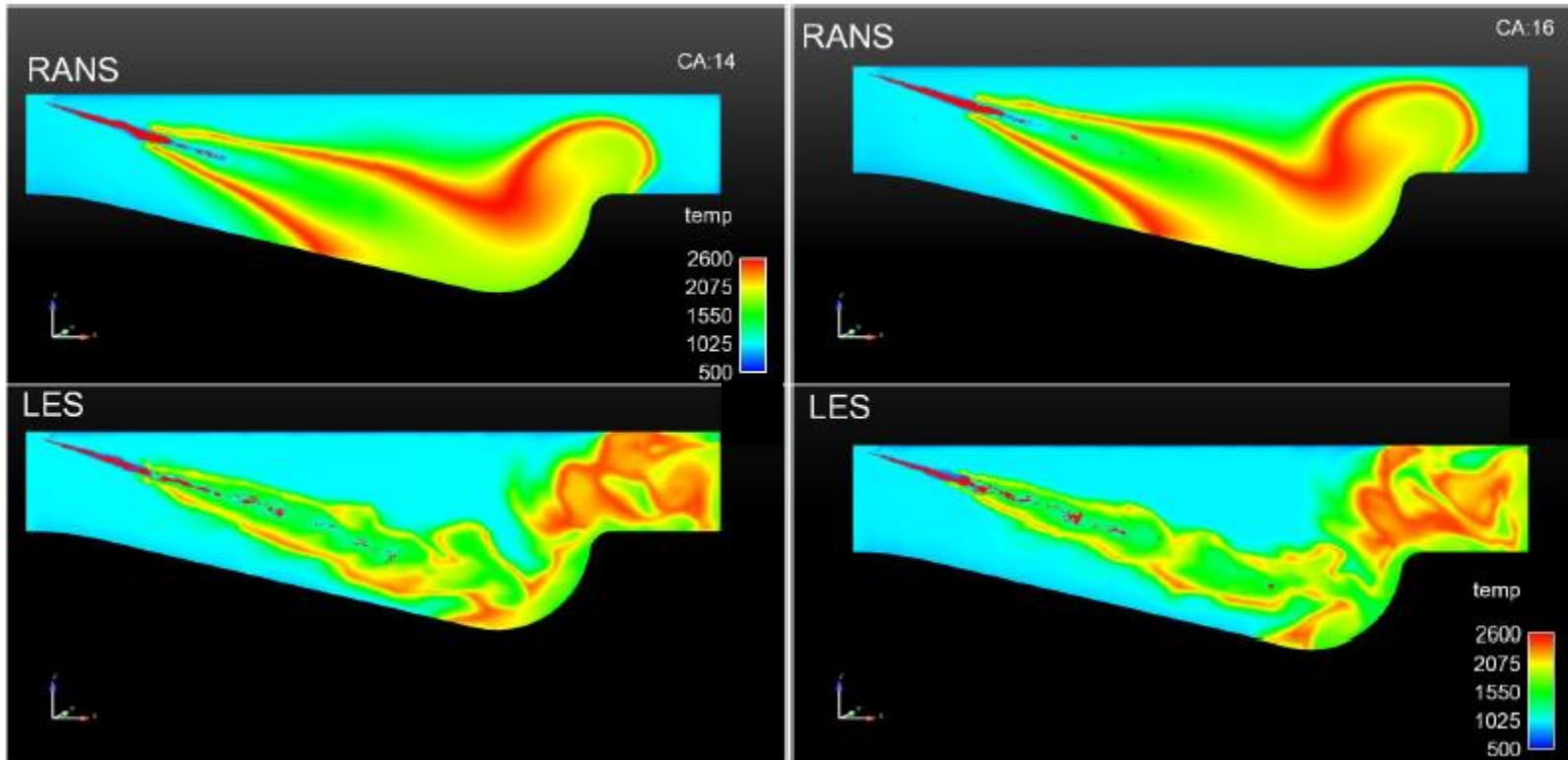
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3D CFD advanced : RANS & LES

RANS : Reynolds-Averaged Navier Stokes

resolves averaged (space & time) turbulent kinetic energy,
Additional turbulence model required



LES : Large eddy simulation

resolves (space & time) turbulent kinetic energy in rough structures
No additional turbulence model required, uses micro structure model

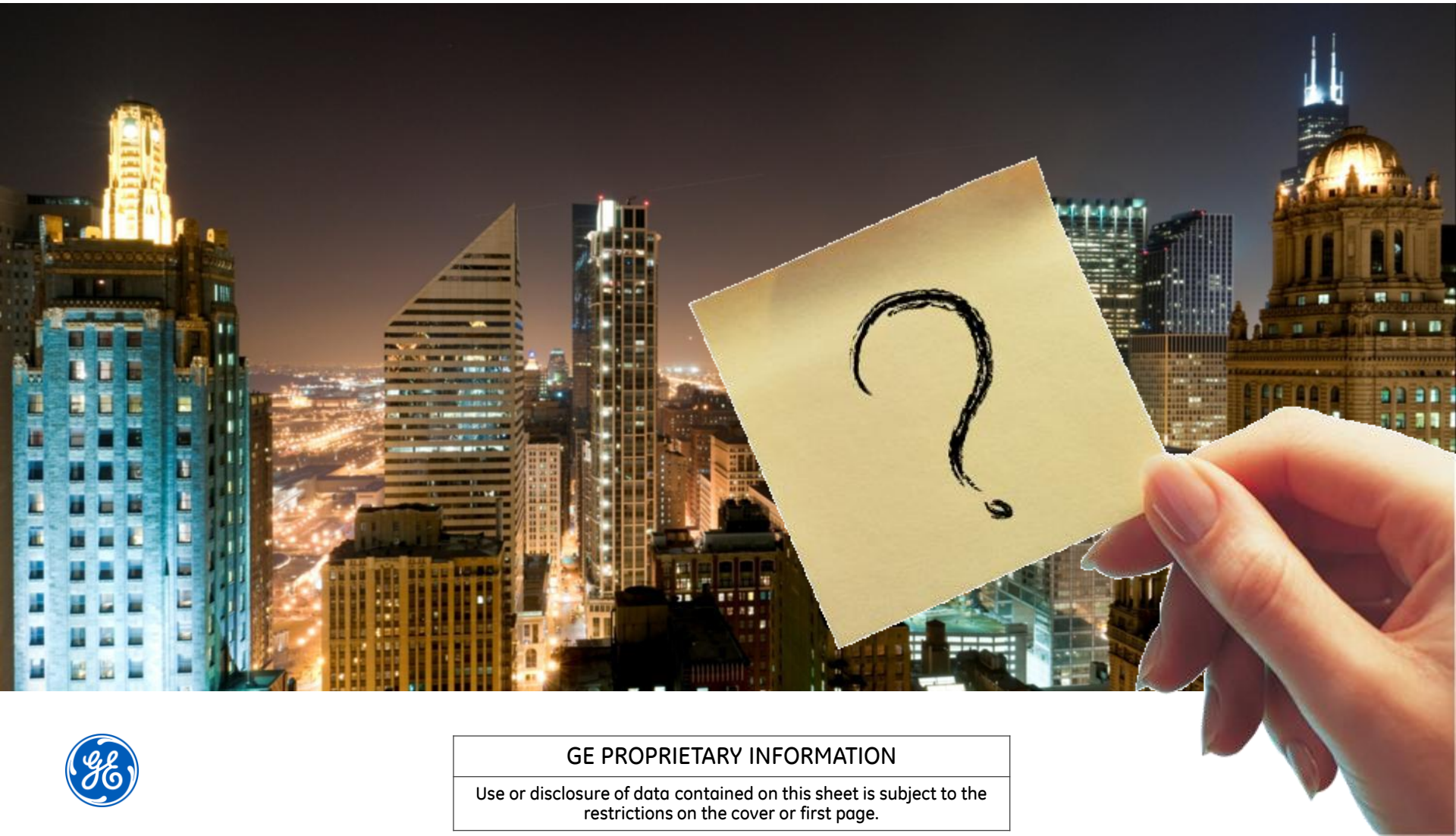


3D CFD + complex chemistry

- Typically conducted for a sector of a given cylinder
- A 3D grid used to section the gaseous/fluid mechanics area of the cylinder
- Grids typically between 10,000 and 1,000,000 cells
- Detailed cylinder geometry, fuel injection, and boundary conditions required
- Individual cell data temporally and spatially resolved available as output
- Time resolution typically on the order of 10 E-9 seconds
- Very high CPU time required (2-14 days)
- High computer storage required (GB)

Performance

How to develop high performance engines



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